

To automate, operate, and scale **Penny Pincher** with near-zero manual overhead, you can deploy a highly specialized team of autonomous AI agents inside your command center. By leveraging an agent orchestration framework (like LangGraph, AutoGen, or CrewAI) hooked directly into your Supabase database, Stripe API, and social channels, this autonomous workforce can handle everything from content creation to fraud monitoring. Here is the blueprint for your autonomous AI employee roster, structured by department.

Department 1: Content & Game Operations

1. The Trivia Alchemist (Content Agent)

- **Role:** Algorithmic Trivia Creator & Fact-Checker.
- **Workflow:** Uses a specialized LLM pipeline to generate 100-question decks for upcoming games. It tags questions by category, formats the multiple-choice arrays, and automatically assigns a dynamic difficulty rating (1 to 100).
- **Safety Loop:** Runs a secondary validation pass to cross-reference facts against trusted knowledge bases, ensuring no duplicate questions or ambiguous answers slip into the PostgreSQL questions table.

2. The Game Director (Scheduling Agent)

- **Role:** Live State Operator.
- **Workflow:** Monitors the Supabase games table. It automatically initializes pending games, schedules the live WebSocket alerts, sets the entry windows, and triggers the transition into **Live Sudden Death Overtime** if the leaderboard detects a tie at Round 100.

Department 2: Growth & Viral Marketing

3. The Hype Machine (Social Media Agent)

- **Role:** Viral Content Creator & Poster.
- **Workflow:** Hooks into your database to pull real-time stats. Right after a game finishes, it automatically generates and posts viral graphics to X, TikTok, and Instagram: *"This Sunday, User @TriviaKing99 turned a 12-cent streak into \$5,550 in cash. Next game starts in 4 hours. Can you pinch the penny?"*
- **Loop:** Continuously monitors app store reviews and social mentions, replying to users with personalized referral links to drive the cost-per-acquisition (CAC) down to zero.

4. The Campaign Commander (User Acquisition Agent)

- **Role:** Smart Ad Optimizer.
- **Workflow:** Analyzes which game modes (Original, Streak Saver, or Milestone) are driving the highest player volume. It dynamically adjusts ad spend across Meta or Google Ads, shifting budget automatically to whichever variant yields the lowest download cost.

Department 3: Risk, Compliance & Finance

5. The Fraud Sentinel (Security Agent)

- **Role:** Real-Time Anti-Cheat & Bot Patrol.
- **Workflow:** Monitors incoming WebSocket payloads during live games. If a user consistently answers difficult Level 80+ questions in under 150 milliseconds (faster than human reading capability), the agent flag-tags the account in `player_game_stats` and sends an alert to the Admin Portal for automated review or temporary wallet suspension.

6. The Ledger Master (Financial Auditor)

- **Role:** Pool Reconciliation & Payout Automation.
- **Workflow:** Runs a strict ACID calculation loop at the conclusion of every game. It verifies that total user micro-debits exactly equal the combined 40% admin revenue pool and 60% player prize pool down to the fraction of a cent. Once verified, it interfaces with the Stripe Connect API to safely move the winnings into the Top 3 players' withdrawable balances.



Data Orchestration Architecture

To keep this agent team unified, you can structure their communication flow through a central system. Here is a visual representation of how your AI agents collaborate inside the command center database:





The Automation Prompt for Claude / CoWork

When you are ready to have your development assistant draft the orchestration logic for these agents, pass it this prompt:

Plaintext

Design an autonomous agent worker using Node.js and a Cron trigger that interfaces with our Supabase setup.

The script must:

1. Scan the 'games' table for games ending within the last 5 minutes.
2. Query the 'player_game_stats' table to extract the top 3 highest scores.
3. Construct a structured JSON payload containing the winners' data and the final total pool size.
4. Format a social media announcement draft designed to be passed to a marketing LLM agent for viral distribution.

This establishes an entirely autonomous business framework, allowing you to focus purely on high-level strategy while your AI workforce manages day-to-day operations and growth scaling.